Uniform bounds for finite Hermite filling functionals

Abstract

We study a finite functional on polynomials of degree less than N, whose spectral nodes are the roots of $2T_N(z/2)+2$ and whose multiplier is an integer Gaussian chirp. Double roots retain first derivatives; the odd simple root has its own value term. We prove that the functional has supremum-norm bound $1+|n|/2$, independently of the level N. If the polynomial vanishes at every node, the bound improves to $(N-1)/N$, independently also of the signed integer n. Tensor products admit a further exact subtraction when only joint grid vanishing is known. The proof uses canonical polynomial remainders, a finite midpoint identity, and an explicit derivative interpolation formula. We also construct a Fourier similarity between a concrete pair Casimir and its polynomial face operator, so the resolvent reproduction identity holds on the full matrix, including nilpotent parts. Regular confluent specialization identifies the Gaussian multiplier with powers of an explicit finite crossing matrix. A marked knot source still requires its separate exact contraction identity; no volume asymptotic is assumed or proved here.

1 Introduction and the main estimate

At a repeated spectral root, a polynomial is not determined by its values. Its derivative can act on a nonzero nilpotent part of an operator. This elementary fact matters when a crossing power is inserted into a finite spectral source: a value-only sum would omit some of the original matrix entries. The purpose of this paper is to retain those entries while bounding the insertion by the supremum of the original polynomial on a real interval.

Fix $N \geq 2$, and write

$$d = N-1, \qquad q = e^{\pi i/N}, \qquad Q = q^2, \qquad \chi_N(z) = 2T_N(z/2)+2.$$

The Chebyshev polynomial is defined by $T_N(\cos\phi) = \cos(N\phi)$; χ_N is monic of degree N. Its distinct double roots are

$$c_a = 2\cos\frac{(2a+1)\pi}{N}, \qquad 0 \leq a < \lfloor N/2 \rfloor.$$

For odd N there is one further root, -2, of multiplicity one. All statements below apply to both parities and to composite levels.

Let $\mathcal{R}_{\chi_N}F$ be the canonical polynomial remainder modulo χ_N, and put

$$\ell_N(F) = [z^{N-1}]\mathcal{R}_{\chi_N}F.$$

For $n \in \mathbb{Z}$, the multiplier f_n is specified at the spectral roots by the local germs

$$f_n(2\cos\phi) = q^{nd^2/4}\exp\left(\frac{inN(\phi-\pi)^2}{4\pi}\right), \qquad 0 < \phi < 2\pi.$$

The formula is symmetric about π and defines a regular germ at $z = -2$. Its Hermite polynomial $\widetilde{f}_n$ is the unique polynomial of degree less than N with those values and first derivatives at the double roots, and the specified value at the odd simple root. Define

$$\mathcal{L}_N^n H = \ell_N(\widetilde{f}_n H).$$

The product is taken before reduction. In particular ℓ_N is not being treated as a multiplicative functional.

Theorem 1.1 (Uniform finite filling bound)**.** *For every complex polynomial H of degree at most d and every signed integer n,*

$$|\mathcal{L}_N^n H| \le (1 + |n|/2)\|H\|_{[-2,2]}. \tag{1.1}$$

If H vanishes at all distinct roots of χ_N, then

$$|\mathcal{L}_N^n H| \le (d/N)\|H\|_{[-2,2]}. \tag{1.2}$$

For a polynomial S in r spectral variables, of degree at most d in each variable, suppose that it vanishes whenever σ_e is a spectral node, with the other variables arbitrary, for each e in a set J. Then

$$\left| \left(\bigotimes_{e=1}^{r} \mathcal{L}_N^{n_e} \right) S \right| \le (d/N)^{|J|} \prod_{e \notin J} (1 + |n_e|/2)\, \|S\|_{[-2,2]^r}. \tag{1.3}$$

Finally, if $\mathcal{D}$ has the same degree bounds and vanishes only on the Cartesian product of all node sets, then

$$\left| \left(\bigotimes_{e=1}^{r} \mathcal{L}_N^{n_e} \right) \mathcal{D} \right| \le \left\{ \prod_e (1 + |n_e|/2) - \prod_e (1/N + |n_e|/2) \right\} \|\mathcal{D}\|_{[-2,2]^r}. \tag{1.4}$$

Here $\|H\|_{[-2,2]} = \max_{-2 \le x \le 2} |H(x)|$, and the cube norm has the analogous meaning. The tensor product denotes successive application in the separate spectral variables. These operations commute because they act in different scalar variables; this says nothing about commutation of matrices at different crossings.

The derivative inequality used below is the classical Bernstein inequality, obtained here from Riesz's finite interpolation formula [2]. The interpolation and polynomial-division tools are standard. The calculations in this paper specify their application to the stated Gaussian Hermite functional, with its normalization and tensor-grid subtraction.

The proof of the theorem occupies Sections 3–8. Its essential estimate is a finite derivative formula with total coefficient mass d. Thus the constant is independent of the number of interpolation nodes, even though the functional retains derivatives at nearly every distinct root. The degree restriction is essential to this argument and will be retained in every application.

The knot-theoretic motivation is the relation between root-of-unity quantum invariants and hyperbolic volume initiated by Kashaev [3] and Murakami–Murakami [4]. The present theorem is a finite estimate. It separates the cost of inserting fixed crossing powers from the harder problem of estimating a shared source polynomial. A short final section states this use as an implication with the exact source identity visible.

2 Polynomial reduction and resolvent reproduction

The value-and-derivative interpolation is the classical construction of Hermite [1]. We record its elementary polynomial form, including repeated roots, because it fixes the coefficient functional used here. Before using the particular spectral roots, we record the algebraic fact behind reduction at repeated roots. It explains which parts of the calculation depend only on the polynomial and which still require the original crossing operator.

Let R be a commutative ring with identity, and let $r_1, \dots, r_m \in R$, with $m \geq 1$. Repetitions are allowed. Put $p(X) = \prod_{j=1}^{m}(X - r_j)$. Given $f \in R[X]$, define $f_0 = f$ and successively define f_j by

$$f_{j-1}(X) = f_{j-1}(r_j) + (X - r_j)f_j(X), \qquad 1 \leq j \leq m.$$

Each f_j is obtained by synthetic division. This is a polynomial operation: it uses no inverse in R, even if two roots coincide. Define

$$\mathcal{R}_p f(X) = \sum_{j=1}^{m} f_{j-1}(r_j) \prod_{i<j}(X - r_i), \qquad \ell_p(f) = [X^{m-1}]\mathcal{R}_p f.$$

Here $[X^k]$ means the coefficient of X^k, and an empty product is one.

Proposition 2.1 (Canonical polynomial remainder)**.** *The polynomial $\mathcal{R}_p f$ is the unique polynomial of degree less than m for which $f - \mathcal{R}_p f$ is a multiple of p. Consequently it is independent of the ordering of the root list, including repeated entries. Both $\mathcal{R}_p$ and ℓ_p are R-linear, and adding a multiple of p to their input leaves their values unchanged. Moreover,*

$$\ell_p(f) = f_{m-1}(r_m).$$

Among additive functionals on $R[X]$, ℓ_p is uniquely specified by the two requirements

$$F(pq) = 0 \quad \textit{for every } q \in R[X], \qquad F(r) = [X^{m-1}]r \quad \textit{whenever } \deg r < m.$$

Proof. Substituting each synthetic-division identity into the preceding one gives $f = pf_m + \mathcal{R}_p f$. Every summand in the displayed remainder has degree at most $m - 1$. If two such remainders existed, their difference would have degree less than m and would be a multiple of the monic polynomial p. A nonzero multiple pq has degree $m + \deg q$: its leading coefficient is the leading coefficient of q, because that of p is one. This argument works even when R has zero divisors. The difference must therefore vanish.

Uniqueness proves independence of root order, linearity and invariance under adding a multiple of p. In the Newton expression for the remainder, only the last summand can contribute to X^{m-1}; its product of linear factors is monic. This proves the formula for ℓ_p. Finally, any additive F with the two stated properties satisfies

$$F(f) = F(pf_m) + F(\mathcal{R}_p f) = [X^{m-1}]\mathcal{R}_p f = \ell_p(f).$$

□

For two equal roots this construction gives

$$\mathcal{R}_{(X-r)^2} f(X) = f(r) + f'(r)(X - r), \qquad \ell_{(X-r)^2}(f) = f'(r).$$

The derivative is the formal polynomial derivative. Thus repeated roots retain derivative information; replacing them by a single evaluation changes the operation.

Proposition 2.2 (Products and repeated-root derivatives)**.** *For any $f, g \in R[X]$,*

$$\mathcal{R}_p(fg) = \mathcal{R}_p((\mathcal{R}_p f)(\mathcal{R}_p g)), \qquad \ell_p(fg) = \ell_p((\mathcal{R}_p f)(\mathcal{R}_p g)).$$

Consequently a finite product may be reduced after each multiplication, with the same final remainder and coefficient functional. This reduced multiplication is associative. If r occurs at least twice among the roots of p, the final remainder retains both $(fg)(r)$ and

$$(fg)'(r) = f'(r)g(r) + f(r)g'(r).$$

For the modulus $(X-r)^2$ its full expression is

$$\mathcal{R}_{(X-r)^2}(fg) = f(r)g(r) + \big(f'(r)g(r) + f(r)g'(r)\big)(X-r).$$

No degree bound on the unreduced factors or their product is required.

Proof. Write $f = \mathcal{R}_p f + pa$ and $g = \mathcal{R}_p g + pb$ using synthetic division. Their product differs from $(\mathcal{R}_p f)(\mathcal{R}_p g)$ by $p\{a\mathcal{R}_p g + b\mathcal{R}_p f + pab\}$. Uniqueness of the remainder proves the first equality; taking its top coefficient proves the second. Repeating the equality proves the finite-product statement. Either parenthesization of a product of three factors reduces the same polynomial fgh, which proves associativity.

When $(X-r)^2$ divides p, the difference between any polynomial and its remainder has both value and formal derivative zero at r. The formal product rule gives the displayed derivative. The explicit degree-one remainder then follows from the double-root formula above. □

The coefficient functional is linear, but it is generally not multiplicative: $\ell_p(fg)$ need not equal $\ell_p(f)\ell_p(g)$. The product must be formed and reduced before its coefficient is taken. At a repeated spectral root the product rule also retains the first derivative of either factor; a list of values alone cannot replace it. This permits a product of arbitrary degree to be formed first and reduced only when the coefficient functional is applied.

There is also a direct algebraic relation between this coefficient functional and the resolvent numerator. It does not require a diagonalization of the operator.

Proposition 2.3 (Reproduction by the polynomial resolvent)**.** *Let $p(X)$ be a split monic polynomial of degree $m \geq 1$ over a commutative algebra A, and let $w \in A$ satisfy $p(w) = 0$. Define V_w by polynomial synthetic division,*

$$V_w(X) = \frac{p(X) - p(w)}{X - w}.$$

The fraction denotes a polynomial, with no inverse assumed in A. Let ℓ_p be the coefficient of X^{m-1} in the canonical remainder modulo p. For every $f \in A[X]$,

$$\ell_p(fV_w) = f(w).$$

Repeated roots of p are allowed.

Proof. Synthetic division gives $(X-w)V_w = p$. The polynomial V_w is monic of degree $m-1$, so $\ell_p(V_w) = 1$. For every polynomial g, the identity $(X-w)gV_w = pg$ and the vanishing of ℓ_p on multiples of p imply

$$\ell_p(XgV_w) = w\ell_p(gV_w).$$

Induction gives $\ell_p(X^j V_w) = w^j$. Linearity proves the claim for arbitrary f, with no restriction on its degree. □

For a matrix application, use the commutative algebra generated by one matrix Ω and scalar matrices. If $p(\Omega) = 0$, then the proposition applies with $w = \Omega$ and gives the full matrix $f(\Omega)$, including its nilpotent part. No diagonalization is required. A concrete Casimir matrix satisfying the required relation is treated in Section 4.

3 Repeated roots, the odd simple root, and the midpoint identity

For completeness, the recurrence $2T_{j+1}(z/2) = z\,2T_j(z/2) - 2T_{j-1}(z/2)$ shows that χ_N is monic of degree N. In the variable ϕ, $\chi_N(2\cos\phi) = 2\cos(N\phi) + 2$ has a double zero whenever $N\phi$ is an odd multiple of π. Away from $\phi = \pi$ the change of variable has nonzero derivative, so these give double roots in z. For odd N, near $\phi = \pi + t$ one has $\chi_N(2\cos\phi) = N^2t^2 + O(t^4)$ and $2\cos\phi + 2 = t^2 + O(t^4)$, giving the simple root -2. The resulting multiplicities add to N, so there are no others.

Write

$$c_a = 2\cos\frac{(2a+1)\pi}{N}, \qquad 0 \le a < \lfloor N/2 \rfloor. \tag{3.1}$$

These are the distinct double roots. The only remaining root is -2, simple, when N is odd. At a double root c, put $h(z) = \chi_N(z)/(z-c)^2$. The Chebyshev differential equation gives

$$h(c) = \frac{N^2}{4-c^2}, \qquad \frac{h'(c)}{h(c)} = \frac{c}{4-c^2}; \qquad \chi_N'(-2) = N^2 \quad (N \text{ odd}). \tag{3.2}$$

Indeed $(4-z^2)\chi_N'' - z\chi_N' + N^2(\chi_N - 2) = 0$. Evaluate it and its derivative at c, using $\chi_N(c) = \chi_N'(c) = 0$, to obtain (3.2).

The formula for the coefficient functional follows directly from partial fractions. For a polynomial F, replace it by its remainder $r = \mathcal{R}_{\chi_N}F$, of degree less than N. The coefficient of z^{-1} in $r(z)/\chi_N(z)$ at infinity is $\ell_{\chi_N}(F)$. At a double root c, the coefficient of $(z-c)^{-1}$ is

$$\left(\frac{r}{h}\right)'(c) = \frac{r'(c)}{h(c)} - \frac{h'(c)r(c)}{h(c)^2}.$$

At the odd simple root it is $r(-2)/\chi_N'(-2)$. Summing these coefficients and using (3.2) gives

$$\mathcal{L}_N^f H = \sum_a \frac{(4-c_a^2)(fH)'(c_a) - c_a f(c_a)H(c_a)}{N^2} + \mathbf{1}_{N \text{ odd}}\frac{f(-2)H(-2)}{N^2}. \tag{3.3}$$

Here f may be a specified holomorphic germ at each root. Its Hermite polynomial is the unique polynomial of degree less than N with those values and first derivatives at the double roots, and its value at the odd simple root. Existence follows because the corresponding linear map between two N-dimensional spaces has zero kernel: a polynomial with all these jets zero is divisible by the monic degree-N polynomial χ_N. We define $\mathcal{L}_N^f H = \ell_{\chi_N}(\widetilde{f}H)$, where $\widetilde{f}$ is that Hermite polynomial. Formula (3.3) therefore applies to every polynomial H, without choosing a global branch for f. Its prime denotes differentiation in z.

Let

$$\phi_k = \frac{(2k+1)\pi}{N}, \quad 0 \le k < N, \qquad P(\phi) = H(2\cos\phi). \tag{3.4}$$

Each double root appears twice in this grid. At the odd simple root, $\phi = \pi$ occurs once, and differentiating the sine supplies exactly its weight in (3.3). Thus (3.3) is equivalently

$$\mathcal{L}_N^f H = -\frac{1}{2}\sum_{k=0}^{N-1} \frac{d}{d\phi}\frac{2\sin\phi\, f(2\cos\phi)H(2\cos\phi)}{N^2}\bigg|_{\phi=\phi_k}. \tag{3.5}$$

There is no omission or doubling of the odd simple node.

For the integer n in Theorem 1.1, the centered Gaussian is

$$f_n(2\cos\phi) = q^{nd^2/4}\exp\left(\frac{inN(\phi-\pi)^2}{4\pi}\right), \qquad 0 \le \phi \le 2\pi. \tag{3.6}$$

Here q^a means $e^{\pi i a/N}$, fixing the original square-root lift even for nonintegral a. In particular

$$|f_n| = 1, \qquad |\partial_\phi f_n| \le \frac{|n|N}{2}, \qquad f_n(c_a) = q^{nd^2/2}(-1)^{an}q^{na(a+1)}. \tag{3.7}$$

The phase in this formula is retained in the functional; it is not replaced by its absolute value. Formula (3.6) is symmetric about π; it has the required holomorphic germ at the odd simple node. The possible endpoint cut at $z = 2$ does not occur at a spectral node and does not enter the finite identity (3.5).

The germ at $z = -2$ is regular. Indeed, with $t = \phi - \pi$ one has $z = -2\cos t = -2 + t^2 + O(t^4)$, so t^2 is a holomorphic function of $z + 2$. At a double root put $h = 2a + 1 < N$. Differentiating the Gaussian and using $dz/d\phi = -2\sin\phi$ gives the exact normal ratio

$$\frac{f_n'(c_a)}{f_n(c_a)} = -\frac{n(N-h)}{2(q^h - q^{-h})}. \tag{3.8}$$

Expanding the square at $\phi = h\pi/N$ gives the value in (3.7); the discarded difference in its exponent is an integer multiple of $2\pi i n$. This calculation is valid for every integer n, positive, negative, or zero, and every $N \ge 2$.

For comparison, let r_N be the degree-less-than-N polynomial with

$$r_N(c_a) = (-1)^a q^{a(a+1)}, \qquad \frac{r_N'(c_a)}{r_N(c_a)} = -\frac{N-1-2a}{2(q^{2a+1} - q^{-2a-1})},$$

and, when $N = 2m + 1$, the value $r_N(-2) = (-1)^m q^{m(m+1)}$ at the simple root. All its nodal values are nonzero, so its class is a unit in $\mathbb{C}[z]/(\chi_N)$. The product rule, including negative powers of a unit, and (3.8) prove

$$\widetilde{f}_n = q^{nd^2/2} r_N^n \pmod{\chi_N}. \tag{3.9}$$

This is an identity of values and normals, not merely an equality of functions on the set of distinct roots. Section 5 identifies $q^{d^2/2} r_N(\Omega)$ with an explicitly defined finite crossing. The norm proof itself requires only the displayed jets.

Expanding (3.5), with primes now in ϕ, gives

$$\mathcal{L}_N^n H = -\frac{1}{N^2}\sum_{k=0}^{N-1}\left(\cos\phi_k f_k P_k + \sin\phi_k f_k' P_k + \sin\phi_k f_k P_k'\right). \tag{3.10}$$

This finite formula is sufficient for every norm below; no interchange of an infinite Fourier series or a contour limit is needed.

4 The local source identity from a finite Fourier transform

We give the local matrix calculation, including its normalization. It supplies an explicit matrix application of the polynomial functional and a homogeneous four-variable representation of its entries. No decomposition into simple modules is needed.

Put $Q = q^2$. In the balanced basis $e_0, \dots, e_d$ of the module of highest weight d and dimension N, the matrices are

$$Ke_i = q^{d-2i}e_i, \qquad Fe_i = (1 - Q^{i+1})e_{i+1}, \qquad \mathcal{E}e_i = (Q^{-i} - 1)e_{i-1}.$$

Here $\mathcal{E} = (q - q^{-1})^2 E$, and a term outside $0, \dots, d$ is zero. One can obtain these matrices from the basis v_i with $Ev_i = [i]_q^2 v_{i-1}$ and $Fv_i = v_{i+1}$ by setting $e_i = v_i/(Q;Q)_i$. Here $[i]_q = (q^i - q^{-i})/(q - q^{-1})$ and $(Q;Q)_i = \prod_{j=1}^{i}(1 - Q^j)$, with the empty product equal to one. All denominators are nonzero because $i < N$. Alternatively the displayed matrices can be taken as the definition of the model. The pair matrices are

$$\begin{aligned} K_{12} &= K \otimes K, \\ F_{12} &= F \otimes I + K^{-1} \otimes F, \\ \mathcal{E}_{12} &= \mathcal{E} \otimes K + I \otimes \mathcal{E}, \\ \Omega &= F_{12}\mathcal{E}_{12} + qK_{12} + q^{-1}K_{12}^{-1}. \end{aligned}$$

On an input $e_u \otimes e_v$, set $U = Q^u, V = Q^v$. Multiplying these displayed matrices gives the diagonal coefficient D, and the two coefficients A_+, A_- that change the first index by $+1, -1$:

$$\begin{aligned} D(U,V) &= 2qU + 2q^{-1}V^{-1} - (q + q^{-1})U/V, \\ A_+(U,V) &= (V^{-1} - 1)(1 - QU), \\ A_-(U,V) &= (Q^{-1}V^{-1} - 1)(1 - U). \end{aligned} \tag{4.1}$$

The second index changes by the opposite amount. The positive shift vanishes when $u = d$ or $v = 0$, and the negative shift vanishes when $u = 0$ or $v = d$. Thus these formulas retain the true integer total and all physical endpoints.

For independent outer face variables x, y, define an operator on polynomials in z by

$$\begin{aligned} (Tg)(z) = &-q^{-1}\left(1 + \frac{q(x+y)}{z} + \frac{q^2xy}{z^2}\right) g(Q^{-1}z) \\ &- q\left(1 + \frac{q^{-1}(x+y)}{z} + \frac{q^{-2}xy}{z^2}\right) g(Qz) \\ &+ \left(\frac{2(x+y)}{z} + \frac{(q+q^{-1})xy}{z^2}\right) g(z). \end{aligned} \tag{4.2}$$

Its denominators cancel on polynomials. Indeed,

$$Tz^j = A_j z^j + B_j(x+y)z^{j-1} + C_j xy z^{j-2},$$

where

$$\begin{aligned} A_j &= -(qQ^j + q^{-1}Q^{-j}), \\ B_j &= 2 - Q^j - Q^{-j}, \\ C_j &= q + q^{-1} - qQ^{-j} - q^{-1}Q^j. \end{aligned}$$

The only potentially negative powers have coefficients $B_0 = C_0 = C_1 = 0$. In particular T preserves the N-dimensional space of polynomials of degree at most d, and it preserves total degree when x, y, z all have degree one.

Proposition 4.1 (The actual Casimir and the face operator)**.** *For $s = 0, \dots, d$, let Ω_s be the restriction of Ω to the ordered basis*

$$e_u \otimes e_{[s-u]_N}, \qquad 0 \le u < N,$$

where $[a]_N$ is the representative of a in $\{0, \dots, N-1\}$. Choose $y \neq 0$, set $x = Q^{s+1}y$, and put

$$b_u = -q^{-1}xQ^{-u}, \qquad \mathsf{E}_{uj} = b_u^j, \qquad G_{uu} = q^{du}.$$

The matrix E evaluates a polynomial at the N distinct points b_u. With T written in the monomial basis,

$$\mathsf{E}T\mathsf{E}^{-1} = -G^{-1}\Omega_s^{\mathsf{T}}G. \tag{4.3}$$

Consequently the full original pair matrix satisfies $\chi_N(\Omega) = 0$, including its nilpotent parts.

Proof. At $z = b_u$, the two physical clocks are

$$-q^{-1}x/b_u = Q^u, \qquad -q^{-1}b_u/y = Q^{s-u}.$$

The coefficients in (4.2) are therefore $-D$, $q^{-1}A_+$, and qA_-. Since $q^d = -q^{-1}$, the positive coefficient is $-q^dA_+$, and the negative one is $-q^{-d}A_-$. Evaluation at $Q^{-1}b_u$ increments the first index, while evaluation at Qb_u decrements it. The endpoint zeros in (4.1) kill every attempted wrap. Thus the coefficient from row u to column r is $-q^{d(r-u)}(\Omega_s)_{ru}$, proving (4.3). The transpose occurs because evaluation acts on functions of the input index, whereas the physical matrix sends input columns to output rows.

The matrix of $-T$ is triangular, with diagonal $t_j + t_j^{-1}$, where $t_j = qQ^j$. Its characteristic polynomial is

$$\prod_{j=0}^{N-1} (X - t_j - t_j^{-1}) = 2T_N(X/2) + 2.$$

To check the identity with multiplicities, substitute $X = w + w^{-1}$ and multiply by w^N. The left side becomes

$$\prod_{j=0}^{N-1} (w - t_j)(w - t_j^{-1}) = (w^N + 1)^2,$$

because both lists are exactly the roots of $t^N = -1$. The right side gives the same Laurent polynomial. Substitution $X = w + w^{-1}$ is injective on polynomials, so the identity follows. Cayley–Hamilton and (4.3) give $\chi_N(\Omega_s) = 0$. The N residue spaces form a direct sum of the full pair space, proving the assertion for Ω. This proves a matrix identity, not only an identity on its eigenvalues. □

The same transform gives the complete local polynomial adapter. For any polynomial f, write

$$f(-T)z^j = \sum_{\alpha,\beta \ge 0,\ \alpha+\beta \le j} M_f(j; \alpha, \beta) x^\alpha y^\beta z^{j-\alpha-\beta}$$

and define

$$H_f(x,b,c,y)=\sum_{j=0}^{d}\sum_{\alpha+\beta\le j}M_f(j;\alpha,\beta)x^{\alpha}b^{j-\alpha-\beta}c^{d-j}y^{\beta}. \tag{4.4}$$

Its exponents are nonnegative and its total degree is d, regardless of the degree of f. Fourier inversion is particularly simple:

$$(\mathsf{E}^{-1})_{jr}=\frac{1}{N}b_r^{-j}.$$

For a polynomial f, (4.3) gives

$$\mathsf{E}f(-T)\mathsf{E}^{-1}=G^{-1}f(\Omega_s)^{\mathsf{T}}G.$$

Use the displayed inverse of E. For $b=b_u, c=b_r$, the result is

$$(f(\Omega_s))_{ru}=\frac{1}{N}q^{-d(r-u)}c^{-d}H_f(x,b,c,y). \tag{4.5}$$

This proves the factor $1/N$, the direction of the shift and the full polynomial identity directly from the original pair matrices. If the input and output have different integer totals, the left side is zero: the modular block is the direct sum of its actual-total blocks. Hence the right side also vanishes at such a grid point.

At even levels $q^{d(p+N)}=-q^{dp}$. The exponent $r-u$ in (4.5) is therefore the canonical integer displacement, not an arbitrarily chosen residue. The endpoint calculation above is what permits the Fourier transform; it does not assume that this phase is periodic. With the coherent face-logarithm lifts, $c=Q^{-(r-u)}b$ converts the prefactor to $N^{-1}(bc)^{-d/2}$, with these coherent choices of logarithms. This rewriting alone does not identify a closed diagram contraction.

This argument holds at every fixed root with $N\ge 2$, including composite levels. It proves the annihilator and the adapter for polynomials of the actual Casimir. The polynomial representing a crossing power and its repeated-node derivatives are established in Section 5. The comparison with a specified knot invariant requires its marked diagram adapter. No derivative with respect to a generic deformation of q is taken here.

5 Crossing values and their repeated-root normals

The Gaussian multiplier was defined by its germs, so the norm theorem is independent of any matrix convention. We now identify it with powers of an explicitly specified finite crossing. This also explains why the first derivatives in the functional are required. The global comparison of a marked diagram with its shared polynomial remains a separate step.

Let $z\in\mathbb{C}^*$ first be non-root-of-unity, write $m=N-1$, and set $\rho_z=z-z^{-1}$. On the basis $v_0,\dots,v_m$ define

$$\begin{aligned} Hv_j&=(m-2j)v_j, & Kv_j&=z^{m-2j}v_j,\\ Ev_j&=[j]_z[m-j+1]_zv_{j-1}, & Fv_j&=v_{j+1}, \end{aligned}\qquad [j]_z=\frac{z^j-z^{-j}}{z-z^{-1}}.$$

As before, endpoint vectors outside the range are zero. Define $[k]_z!=\prod_{r=1}^{k}[r]_z$, with $[0]_z!=1$, and put

$$\mathscr{R}_z=z^{H\otimes H/2}\sum_{k=0}^{m}\alpha_kE^k\otimes F^k,\qquad \alpha_k=\frac{z^{k(k-1)/2}\rho_z^k}{[k]_z!},\qquad B_z=\mathsf{P}\mathscr{R}_z,$$

where $\mathsf{P}(v \otimes w) = w \otimes v$. Half-integral powers are read on a fixed local square-root branch; at $z = q$ this is the branch $q^a = \exp(\pi i a/N)$. The pair operators are

$$E_{12} = E \otimes K + I \otimes E, \quad F_{12} = F \otimes I + K^{-1} \otimes F, \quad K_{12} = K \otimes K,$$

and their Casimir is $\Omega_z = \rho_z^2 F_{12} E_{12} + z K_{12} + z^{-1} K_{12}^{-1}$. The specialization of this matrix in the balanced basis is the Ω of Section 4.

Proposition 5.1 (The finite crossing polynomial)**.** *At $z = q$, the displayed finite crossing satisfies*

$$B_q = q^{d^2/2} r_N(\Omega), \qquad B_q^n = \widetilde{f}_n(\Omega) \quad (n \in \mathbb{Z}),$$

where r_N and $\widetilde{f}_n$ are the polynomials defined in (3.9). These are full matrix identities.

Proof. We first verify the intertwining that will be used to find the scalar on each generic summand. The displayed matrices satisfy $KEK^{-1} = z^2 E$, $KFK^{-1} = z^{-2} F$, and $[E, F] = (K - K^{-1})/\rho_z$. Induction gives

$$[E, F^k] = \frac{[k]_z}{\rho_z} F^{k-1} (z^{-(k-1)} K - z^{k-1} K^{-1}),$$
$$[E^k, F] = \frac{[k]_z}{\rho_z} E^{k-1} (z^{k-1} K - z^{-(k-1)} K^{-1}).$$

Write $A = \sum_k \alpha_k E^k \otimes F^k$ and $D = z^{H \otimes H/2}$. Moving D through the pair generators reduces the two required intertwining identities to

$$A(E \otimes K + I \otimes E) = (I \otimes E + E \otimes K^{-1}) A,$$
$$A(F \otimes I + K^{-1} \otimes F) = (K \otimes F + F \otimes I) A.$$

For $0 \le k < m$, the coefficient in the first difference multiplying $E^{k+1} \otimes F^k (K - z^{2k} K^{-1})$ is $\alpha_k - \alpha_{k+1} [k+1]_z z^{-k}/\rho_z$. For the second, the coefficient multiplying $E^k (z^k K - z^{-k} K^{-1}) \otimes F^{k+1}$ is $\alpha_{k+1} [k+1]_z/\rho_z - \alpha_k z^k$. Both vanish by the displayed formula for α_k; the terminal terms vanish because $E^N = F^N = 0$. Intertwining with K_{12} follows from the weights. Hence B_z commutes with E_{12}, F_{12}, K_{12}. The same commutation relations for these pair generators give $[\Omega_z, E_{12}] = [\Omega_z, F_{12}] = [\Omega_z, K_{12}] = 0$ by substitution into the displayed formula for Ω_z.

For $0 \le k \le m$, seek a highest vector

$$h_k = \sum_{a=0}^{k} b_a v_a \otimes v_{k-a}, \qquad b_0 = 1.$$

The equation $E_{12} h_k = 0$ gives

$$\frac{b_{a+1}}{b_a} = -\frac{[k-a]_z [m-k+a+1]_z}{[a+1]_z [m-a]_z \, z^{m-2k+2a+2}}, \qquad b_k = (-1)^k z^{-k(m-k+1)}.$$

The last formula follows by multiplying the ratios: the quantum-integer factors cancel in reverse order. The highest weight is $2m - 2k$. For a highest vector of weight λ, direct induction gives

$$E_{12} F_{12}^r h = [r]_z [\lambda - r + 1]_z F_{12}^{r-1} h.$$

It follows that h_k generates a module of dimension $2m-2k+1$: none of these vectors can vanish before that length. The next vector, if nonzero, would be a highest vector of negative integral weight. For a negative highest weight λ, both $[r]_z$ and $[\lambda-r+1]_z$ are nonzero for every $r \geq 1$, since z is not a root of unity. Induction in the same identity would force every lowering to be nonzero, contrary to nilpotence of F_{12}. The module is irreducible because any nonzero invariant subspace contains a weight vector: the distinct eigenvalues of K_{12} allow its weight projections to be written as polynomials in K_{12}. That vector can be raised to h_k and then lowered through the whole module.

On this module the Casimir scalar is

$$\eta_k(z) = z^{2N-1-2k} + z^{-2N+1+2k}.$$

For generic z these scalars are distinct, and the displayed dimensions sum to N^2. Thus the modules form a direct sum of the full pair space. Since B_z commutes with the generators, it is scalar on each one: it preserves the one-dimensional highest-weight line, and commutation with F_{12} fixes the same scalar on every lowering. To compute its scalar on h_k, examine the coefficient of $v_0 \otimes v_k$ in $\mathsf{P}\mathscr{R}_z h_k$. Only the zero-th term of the sum defining $\mathscr{R}_z$, acting on $v_k \otimes v_0$, can contribute. Therefore the scalar is

$$\lambda_k(z) = (-1)^k z^{m^2/2-k(2N-1-k)}.$$

Ordinary interpolation at the distinct generic scalars supplies a polynomial p_z of degree less than N such that $B_z = p_z(\Omega_z)$.

It remains to specialize this identity without discarding the normal part. At $z = q$ the nodes k and $l = N-1-k$ collide, and their crossing values agree. Put $h = 2k+1 < N$ and $A_h = q^h - q^{-h} \neq 0$. Differentiating with respect to $\log z$ gives

$$\begin{aligned} \partial_{\log z}(\eta_k - \eta_l)|_q &= -2NA_h, \\ \partial_{\log z}(\lambda_k - \lambda_l)|_q &= \lambda_k(q)N(N-h). \end{aligned}$$

Consequently the limiting divided difference is

$$\lim_{z\to q} \frac{\lambda_k(z) - \lambda_l(z)}{\eta_k(z) - \eta_l(z)} = -\frac{\lambda_k(q)(N-h)}{2(q^h - q^{-h})}.$$

Replace each colliding pair of interpolation rows by its first row and its divided-difference row. The coefficient matrix tends to the confluent Vandermonde matrix for the double roots c_a and the odd simple root when present. It is invertible: a polynomial of degree less than N in its kernel has all the prescribed zero jets and is therefore divisible by χ_N. Thus the coefficients of p_z are regular at q, and their limits are the specified Hermite data. The value is $\lambda_k(q) = q^{d^2/2}(-1)^k q^{k(k+1)}$.

Every entry of B_z is regular there as well, since $[j]_q! \neq 0$ for $j < N$. Specializing the matrix identity proves $B_q = q^{d^2/2} r_N(\Omega)$ with its nilpotent part included. The nodal values of r_N are nonzero, so its class modulo χ_N is invertible. An inverse is obtained by prescribing the jets $r_N(c)^{-1}$ and $-r_N'(c)/r_N(c)^2$ at each double root, and the inverse value at the odd simple root. The product has the same jets as one, so is one modulo χ_N. Now (3.9), together with $\chi_N(\Omega) = 0$, proves the assertion for every signed power. □

The specialization argument uses a generic parameter to establish a finite matrix identity, not to define a transverse derivative of the knot invariant. Its derivative is used only to recover the coefficient of the repeated spectral root. No generic projector or generic direct sum is specialized at the root.

6 An elementary exact derivative inequality

For every complex trigonometric polynomial P of degree at most $d \geq 1$, set $\theta_j = (2j-1)\pi/(2d)$, $1 \leq j \leq 2d$. The classical Riesz interpolation formula [2] is

$$P'(x) = \sum_{j=1}^{2d} \frac{(-1)^{j+1}}{4d \sin^2(\theta_j/2)} P(x + \theta_j). \tag{6.1}$$

For completeness, invert the finite Fourier matrix on modes $-d+1, \dots, d$. If $z = e^{-i\theta_j}$, so $z^{2d} = -1$, differentiating the finite geometric sum gives its weight

$$\frac{1}{2d} \sum_{\ell=-d+1}^{d} i\ell e^{-i\ell\theta_j} = \frac{iz^{-d+1}}{d(1-z)^2} = \frac{(-1)^{j+1}}{4d \sin^2(\theta_j/2)}. \tag{6.2}$$

The extra mode $-d$ has samples opposite to those of d, and its derivative at zero is opposite too. Thus (6.1) holds on all modes $-d, \dots, d$; translation gives arbitrary x.

The elementary sine product $\prod_{j=0}^{2d-1} \sin(x + j\pi/(2d)) = C_d \sin(2dx)$, with $C_d \neq 0$, yields after two logarithmic derivatives and substitution $x = \pi/(4d)$

$$\sum_{j=1}^{2d} \frac{1}{4d \sin^2(\theta_j/2)} = d. \tag{6.3}$$

Taking absolute values in (6.1) therefore proves

$$\|P'\|_{\mathbb{R}} \leq d \|P\|_{\mathbb{R}}. \tag{6.4}$$

The argument works without real coefficients or positivity.

The same finite identity and triangle inequality hold when the coefficients lie in any finite-dimensional normed space. Thus (6.4), and consequently (1.1)–(1.4), hold for such vector-valued polynomials without a dimension factor. The intended application here remains the original shared scalar coefficient of Section 9; no change of the source norm is required.

7 Full and vanishing-node bounds

Since H has degree at most d, $P(\phi) = H(2\cos\phi)$ has trigonometric degree at most d, and its supremum is exactly $\|H\|_{[-2,2]}$. Equations (3.7), (3.10), and (6.4) give

$$|\mathcal{L}_N^n H| \leq \frac{N}{N^2} \left(1 + \frac{|n|N}{2} + d\right) \|H\| = \left(1 + \frac{|n|}{2}\right) \|H\|, \tag{7.1}$$

because $d + 1 = N$. This proves (1.1).

If H vanishes at every node, the first two terms of (3.10) vanish exactly. Only the derivative of the original numerator remains. Applying (6.4) to that term gives (1.2), uniformly in every signed n, including zero.

Node vanishing does not make the functional zero. For any double root c_a, the polynomial

$$H_a(z) = \frac{\chi_N(z)}{z - c_a}$$

has degree d and vanishes at every spectral node, including the odd simple node. At c_a its derivative is $h(c_a)$, and every other term in (3.3) is zero. Hence

$$\mathcal{L}_N^n H_a = f_n(c_a), \qquad |\mathcal{L}_N^n H_a| = 1. \tag{7.2}$$

The improvement in the norm is therefore a bound on the surviving normal contribution, not a claim that all values determine the answer.

8 Shared polynomials and the unsplit complementary source

The other finite functionals preserve each partial node-vanishing condition. Applying (1.1) and (1.2) successively proves (1.3). More explicitly, after the other variables are fixed on their real intervals, each application leaves a polynomial of the same degree bound in the remaining variables. A vanishing condition in a different coordinate is preserved by linearity. The same finite Riesz identity controls derivatives with values in the remaining finite-dimensional polynomial space, with its real-cube supremum norm; there is no dimension factor.

For clarity, (1.4) needs only joint grid vanishing, not vanishing on every node hyperplane. In each coordinate (3.10) is a sum of an evaluation term and a term differentiating P. Before summing its N grid entries, the two coefficient bounds are $1 + |n_e|N/2$ and 1, respectively. For a subset A of differentiated variables, iterating (6.4) gives

$$\|\partial_A \mathcal{D}(2\cos\boldsymbol{\phi})\| \le d^{|A|} \|\mathcal{D}\|_{[-2,2]^r}. \tag{8.1}$$

The term $A = \varnothing$ vanishes pointwise at the tensor grid. Every other term is bounded by $\prod_{e\in A}(d/N)\prod_{e\notin A}(1/N + |n_e|/2)\|\mathcal{D}\|$. Adding the nonempty subsets proves (1.4).

The joint condition arises naturally from a product of local factors. Suppose P_e and C_e have identical values at every spectral node in their e-th variable, as polynomials in all shared auxiliary variables. Then $\prod_e P_e - \prod_e C_e$ vanishes on the full spectral product grid. Any fixed linear extraction of the auxiliary variables preserves this identity. Thus (1.4) applies after that shared extraction, provided the degree in each spectral variable is at most d. This argument does not require vanishing on each node hyperplane.

For example, if $N = 3$ then $\chi_3(z) = (z-1)^2(z+2)$. The polynomial $(z-1)(z+2)$ vanishes at both distinct nodes but has nonzero first derivative at the double root. In several variables a sum of products with different such factors can vanish on the product grid without sharing a factor in any single coordinate. The subtraction of the all-evaluation term in (1.4) remains valid in that case.

9 Application after a shared coefficient extraction

Let $L_1, \dots, L_s$ be auxiliary variables and let $\mathcal{C}$ be a fixed coefficient extraction in those variables, possibly after some have been set to zero. Suppose the local spectral factors $P_e(L; \sigma_e)$ have degree at most d in σ_e. Then

$$S_N(\boldsymbol{\sigma}) = \mathcal{C}\left(\prod_e P_e(L; \sigma_e)\right)$$

has the same separate degree bounds. Theorem 1.1 applies to this shared polynomial itself. It does not replace the extraction by independent maxima of its local factors.

Corollary 9.1 (Exact-source implication)**.** *Assume a scalar contraction $Z_N(\boldsymbol{n})$ has the exact identity*

$$Z_N(\boldsymbol{n}) = \left(\bigotimes_e \mathcal{L}_N^{n_e}\right) S_N. \tag{9.1}$$

Let C_N be any other shared polynomial with the same separate degree bounds, and let $Z_{N,C} = (\bigotimes_e \mathcal{L}_N^{n_e}) C_N$. Then

$$|Z_N - Z_{N,C}| \le \prod_e (1 + |n_e|/2)\, \|S_N - C_N\|_{[-2,2]^r}.$$

If $S_N - C_N$ vanishes on the product node grid, the smaller constant in (1.4) applies.

Proof. The finite functional is linear. Subtract (9.1) and the definition of $Z_{N,C}$ before applying (1.3) with $J = \varnothing$, or (1.4). This preserves the full difference, including every source sector and all shared coefficient demands. □

For the intended Morse-diagram application, the local factor is the homogeneous representation of

$$V_N(\sigma; \Omega) = \frac{\chi_N(\sigma) - \chi_N(\Omega)}{\sigma - \Omega},$$

where the fraction is polynomial division. The Fourier calculation in Section 4 proves the local adapter and the annihilator needed by resolvent reproduction. Proposition 5.1 also identifies the defined finite crossing, including its repeated-root normals. A companion exact-source theorem must compare the diagram conventions with these matrices and prove that the original marked tensor contraction has the stated shared coefficient with its actual cups, caps, face grids, and framing scalar. Those are the hypotheses of (9.1), not consequences of a norm bound. The conditional formulation permits this finite paper to be used with any source for which that exact identity has been established.

Here is the resulting asymptotic implication, recorded to make clear which estimate an application must supply. It is an elementary consequence of a strict remainder gap, not an additional source theorem.

Proposition 9.2 (Transfer of a strict remainder gap)**.** *Fix the number of variables and the integer tuple $\boldsymbol{n}$. Suppose (9.1) holds and, for some real Φ and $\eta > 0$,*

$$\|S_N - C_N\|_{[-2,2]^r} \le \exp\big(N(\Phi - \eta) + o(N)\big).$$

Suppose separately that

$$Z_{N,C} = a_N e^{NA}, \qquad a_N \neq 0 \text{ eventually}, \qquad \frac{1}{N}\log|a_N| \longrightarrow 0, \qquad \operatorname{Re} A > \Phi - \eta.$$

Then Z_N is eventually nonzero and

$$\frac{1}{N}\log|Z_N| \longrightarrow \operatorname{Re} A.$$

Multiplication by a specified scalar sequence u_N with $u_N \neq 0$ eventually and $N^{-1}\log|u_N| \to 0$ preserves the conclusion.

Proof. The filling constant in Corollary 9.1 is independent of N. Let $b = \Phi - \eta$ and choose $\epsilon > 0$ with $b + 2\epsilon < \operatorname{Re} A$. For all sufficiently large N,

$$|Z_N - Z_{N,C}| \leq e^{N(b+\epsilon)}, \qquad |Z_{N,C}| \geq e^{N(\operatorname{Re} A - \epsilon)}.$$

Their ratio is at most $e^{-N(\operatorname{Re} A - b - 2\epsilon)}$. In particular it is eventually below $1/2$. The triangle and reverse triangle inequalities give

$$\tfrac{1}{2}|Z_{N,C}| \leq |Z_N| \leq \tfrac{3}{2}|Z_{N,C}|.$$

Taking logarithms and dividing by N proves the limit and eventual nonvanishing. The last assertion follows by adding $N^{-1}\log|u_N|$. □

A bound for the full source at the larger rate Φ would not satisfy this proposition when $\operatorname{Re} A < \Phi$. Nor does a leading term for C_N estimate the omitted source. These two inputs have to be proved separately for the same fixed diagram and normalization. Identifying $2\pi \operatorname{Re} A$ with a particular hyperbolic volume requires a further geometric argument; it is not part of the finite functional theorem.

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